QP CODE: 25900650

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU-UGP(HONOURS) REGULAR EXAMINATION

OCTOBER 2025

MG3DSCSTA202 - DATA ANALYSIS IN INFERENTIAL STATISTICS USING R/PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum marks: 50**

SCHEME OF VALUATION

PART A

Short Answer Questions

Answer any seven questions. Each question carries two marks.

1. Definition of conditional probability - 1 Mark,
   P(X=0|Y=0) =0.5 -1 Mark
   [Definition of conditional probability only - 2 Marks
   Maximum =2 marks]
2. Property of probability mass function, Σ f(x)=1 – 1 Mark
   Any reasonable attempt to determine C – 1 Mark
   P (x, y) = c x^2y; c=1/28
   P (x, y) =cx2y; c=1/36
3. Statement Z= $(X-\mu)/\sigma \sim N(0,1)$ – 1 Mark
   Answer= 20% - 1 Mark
4. Any two applications -2 Mark
5. No, F distribution cannot take negative values- 2Mark
   Range of F Distribution is $(0, \infty)$
6. One advantage – 1 Marks (eg: Moment estimates are asymptotically unbiased)
   One disadvantage – 1 Mark (eg: If moment does not exist, method of moment is not applicable)
7. Definition/Example – 2 Mark
8. Definition/Example -2 Mark
9. Any two assumptions for Chi-square test for homogeneity/General assumptions of Chi-square test – 2 Mark
10. Statistic- 2 Mark
    t= $(\frac{X-\mu)\sqrt{(n-1)}}{s} \sim t\ distribution\ with\ (n-1)\ degrees\ of\ freedom$

PART B

Short Essay Questions

Answer any four questions. Each question carries six marks.

11. $f_1(x)$= 2x+3/21; x=1, 2, 3 – 2 Mark
    $f_2(y)$ = 6+3y/21; x=1,2 – 2 Marks

Condition for independence – 1Mark
Checking independence – 1 Mark
Condition for independence only – 2 Mark

12. U+V $\sim\chi 2$ (2) Distribution
(Chi square distribution with 2 degrees of freedom – 3 Marks
MGF=$(1-2t)^{-1}$ -1 Mark
Mean = 2 – 1 Mark
Variance = 4 – 1 Mark
[Specify Chi square pdf, mean, variance and mgf – 4 Marks (1 mark each)
13. Definition of sampling distribution - 3 Mark
Define any statistic – 2 Mark
Sampling distribution– 1 Mark
14. Definition of consistency – 2 Mark
Sufficient conditions – 2 Marks each
15. H_0: P= 18/38, H_1: P≠18/38 - 1 Mark each
t =$\frac{p-po}{\sqrt{p_0 q_0}/n} \sim N(0,1)$ – 1 Marks
Answer=1.98 – 1 Mark
Table value= 1.96 - 1 Mark
Conclusion -1 Mark
Reject H_0; This suggests that the wheel may not be fair in terms of red outcomes.
[4 marks shall be awarded for any reasonable attempt.]
16. Any reasonable brief explanation of ANOVA – 2 Marks
MSC=SSC/K-1 =61.665- 1/2 Marks
MSE=SSE/N-K=7.833 -1/2 Marks
F=MSC/MSE = 7.87-1 Marks
Table value = 3.88-1 Marks
Conclusion – 1 Marks
Reject the hypothesis, there is a significant difference in the average score

PART C

Essay Questions

Answer any one question. Each question carries 12 marks.

[Four marks shall be awarded for any reasonable attempt, if the solution does not qualify for the step marks mentioned.]

17. Definition of joint probability mass function – 3 Marks
Two properties -3 Marks
Correct pmf = 4 Marks
f (0,0) =0, f (0,1) =2/6, f (1,0) = 2/6, f (1,1) =2/6
Correct distribution function = 2 Marks

18. Explanation of any one large sample test for population mean-3 Mark
Define H_0 and H_1- 2 Mark
Correct test statistic= 2 Mark
Z= $\frac{(X-\mu)\sqrt{n}}{\sigma} \sim N(0,1)$

Answer= -1.2 -2 Marks
Table value= 1.96- 2 Mark
Conclusion -1 Mark
Accept the hypothesis